Discovery of Diagnostic and Prognostic long non-coding RNA Biomarkers in Patients with Colorectal Cancer Using Machine Learning Algorithms.

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ABSTRACT

Table S1. Statistical baseline information of clinical features in the training and testing group.

Item	Group		p-value
	Training (n=300)	Testing (n=128)	
Overall survival (days)	880 (±752)	915 (±864)	0.689
Age (years)	66.1 (±13.3)	68.1 (±12.3)	0.120
Preoperative CEA (ng/ml)	33.6 (±197)	50.4 (±218)	0.555
Status			1.000
censored	234 (78.0%)	100 (78.1%)	
dead	66 (22.0%)	28 (21.9%)	
Gender			0.449
male	156 (52.0%)	72 (56.2%)	
female	144 (48.0%)	56 (43.8%)	
Stage			0.988
i	51 (17.0%)	22 (17.2%)	
ii	117 (39.0%)	51 (39.8%)	
iii	90 (30.0%)	36 (28.1%)	
iv	42 (14.0%)	19 (14.8%)	
Lymphatic invasion			0.909
no	167 (55.7%)	69 (53.9%)	
yes	106 (35.3%)	45 (35.2%)	
missing	27 (9.0%)	14 (10.9%)	
Perineural invasion			0.513
no	98 (32.7%)	31 (24.2%)	
yes	35 (11.7%)	8 (6.2%)	
missing	167 (55.7%)	89 (69.5%)	
Venous invasion			
no	201 (67.0%)	82 (64.1%)	0.685
yes	61 (20.3%)	28 (21.9%)	
missing	38 (12.7%)	18 (14.1%)	

Table S2. Statistical information of lncRNA features in patient's group.

	Survival Status		
Item	**Censored (n=334)**	**Dead (n=94)**	**P-value**
Overall survival (days)	940 (±802)	717 (±703)	0.009
Relative Risk Score	1.43 (±1.98)	4.92 (±5.88)	<0.001
Risk Group			
high	141 (42.2%)	73 (77.7%)	<0.001
low	193 (57.8%)	21 (22.3%)	
Group			1.000
Testing	100 (29.9%)	28 (29.8%)	
Training	234 (70.1%)	66 (70.2%)	
LncRNA Expression			
RP11-440D17.3	-0.203 (±2.29)	0.158 (±2.39)	0.194
EIF3J-AS1	0.876 (±0.564)	1.04 (±0.567)	0.012
TNRC6C-AS1	0.432 (±0.906)	0.705 (±0.917)	0.012
MIR210HG	1.26 (±1.24)	1.63 (±1.23)	0.011
AC113189.5	1.35 (±0.727)	1.66 (±0.913)	0.003
LINC00261	1.48 (±2.18)	0.803 (±2.07)	0.006
CTB-25B13.12	2.15 (±0.708)	2.45 (±0.639)	<0.001
AC114730.3	1.20 (±1.43)	0.788 (±1.89)	0.053

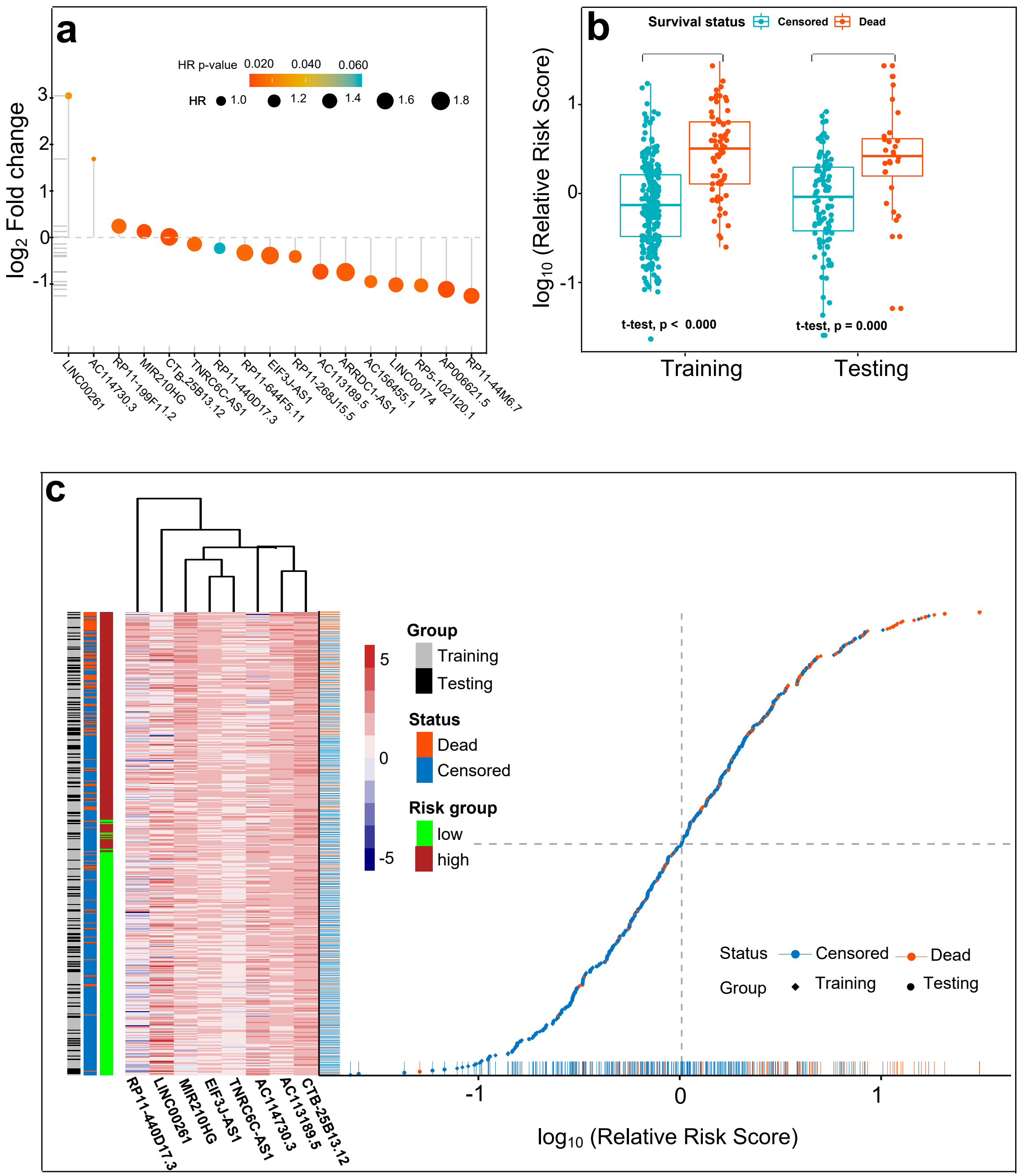

Figure S1. (a) The features of the 17 candidate lncRNAs : HR was depicted as the size of dots with corresponding color annotated p-value, and $\log_2$ FC as the y-axis. Most of the downregulated lncRNAs associated with increased HR, while LINC00261, a putative negative regulator of cell growth through promoting differentiation and apoptosis was significantly upregulated with decreased HR. (b) Boxplot of $\log_{10}(RelativeRiskScore)$ in groups: $\log_{10}(RelativeRiskScore)$ were significant higher in the patient suffering from death of CRC in both groups. (c) The patients' relative risk score distribution and lncRNA expression heatmap: The patients were evenly partitioned in training and testing group as the grey and black legend bar shown. Relative Risk Score stratified patients into high and low risk group shown as dark-red and green bar. The death experienced and high risked patients mainly clustered in the upright region on the graph.

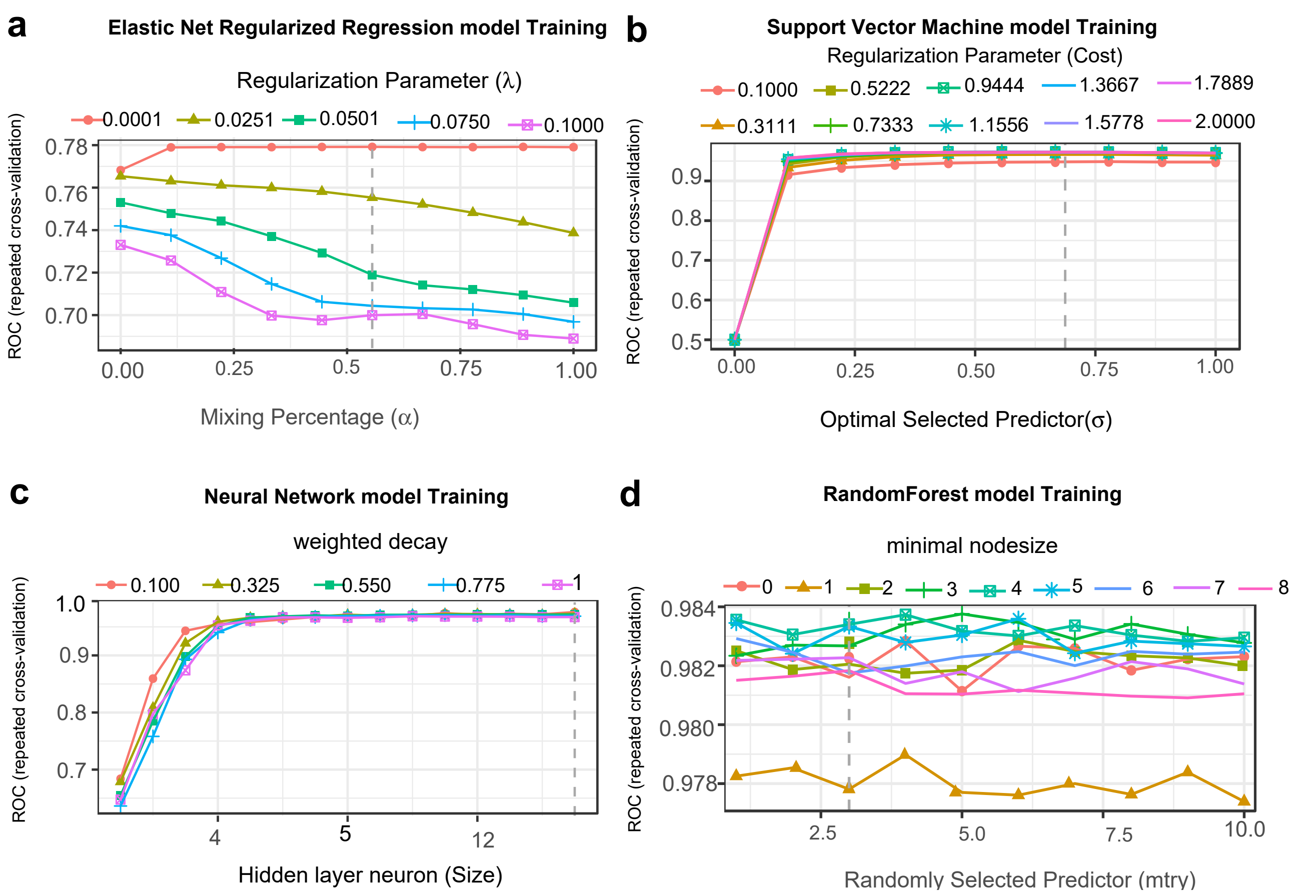

Figure S2. Repeated cross-validated ROC and optimal tuning hyperparameter in each classification model: Each dashed vertical line indicating where the optimal point located. (a) Elastic net regularized regression model ($\alpha = 0.556$, $\lambda = 1 \times 10^{-4}$). (b) Support vector machine model ($Cost = 0.944$, $\sigma = 0.667$), (c) Neural network model ($Size = 15$, $decay = 0.100$), (d) Random forest model ($mtry = 3$, $minimal\ nodesize = 5$)